

P530/2

Biology

Paper 2(THEORY)

2 $\frac{1}{2}$ hours



Apex Examination Council

MOCK EXAMINATIONS 2025

Uganda Advanced certificate of education

BIOLOGY

Paper 2(THEORY)

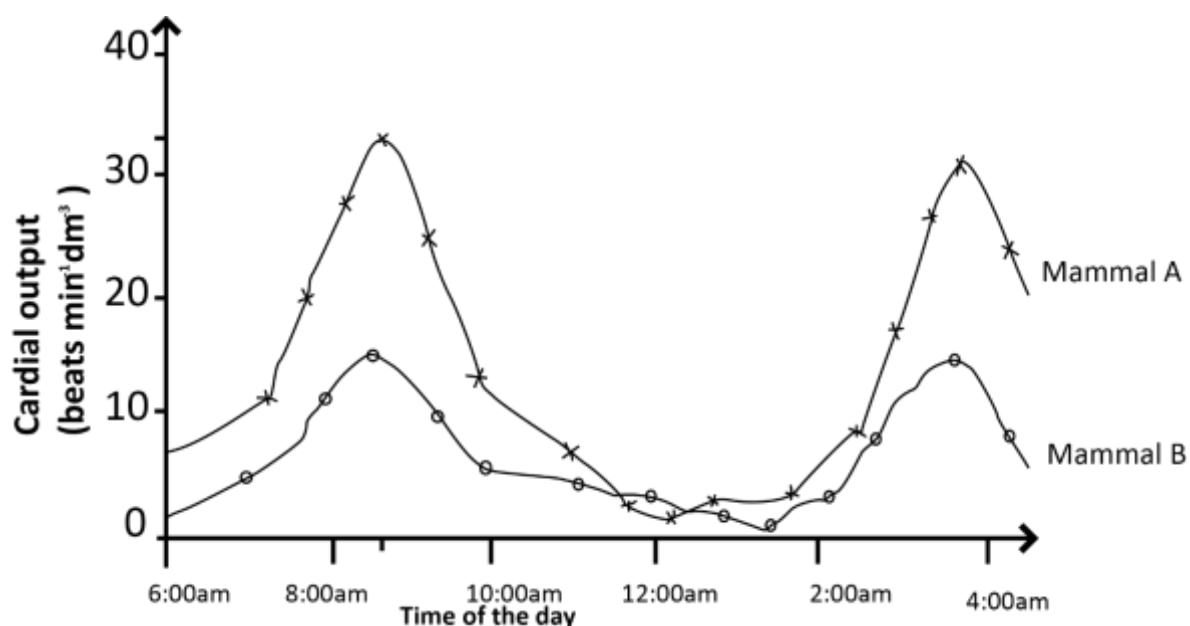
2 hours 30 Minutes

INSTRUCTIONS TO CANDIDATES:

- This paper consists of six questions
- Answer question one in section A plus three others from section B.
- Begin a new question on a fresh page.
- Candidates are advised to read the questions carefully, organize their answers and present them precisely and logically, illustrating with well labelled diagrams where necessary.

SECTION A (40 MARKS)

1. The figure shows the changes in the cardiac output of two individual Mammals A and B of different sizes, determined from 6:00a.m up to 4:00p.m in the evening when each mammal was given a hot drink.



- a). Compare the cardiac output of both mammals.(07 marks)
 - b). Explain the variation of the cardiac output of mammal A.(08marks)
- 2.Comment on the difference in the cardiac output both mammals.(06marks)
- a). Suggest factors that are likely to affect the cardiac output of a mammal.(04marks)
 - b). Suggest and explain how each of the following can affect the cardiac output of mammal A.
 - (i) exercise (08 marks)
 - (ii) altitude (04 marks)
- c).How is the cardiac output regulated in man. (03 marks)

SECTION B (60marks)

3. a). What is meant by reciprocal innervation? (03 marks)
- b). Differentiate between smooth and cardiac muscles. (07 marks)
- c). Explain how a skeletal muscle shortens when stimulated. (10 marks)
4. (a) Distinguish between transcription and translation. (05marks)
- (b) Explain how translation directs the synthesis of haemoglobin in a cell. (12 marks)
- (c) What is the significance of translation in organisms? (03 marks)
5. a) How is the cell wall adapted to its functions in higher plants. (09 marks)
- (b) Explain how each of the following occurs across the cell membrane:
- (i) active transport. (06 marks)
- (ii) endocytosis (05 marks)
6. (a) How does each of the following support evolution?
- (I) distribution of phosphagens. (06 marks)
- (ii) homologous structures (06 marks)
- (b) Explain how the environment may influence the process of natural selection. (08 marks)
7. a) Describe the;
- (I) characteristics of bacteria that make them abundant in nature (05marks)
- (ii) role of bacteria in the Nitrogen cycle (07 marks)
- (b) Explain how humans interfere with the nitrogen cycle. (08 marks)

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